Triangle Types and the Extension Gap: Towards Decidability of $\mathcal{ALCI}_{RCC5}$

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Status (April 2026). This paper presents a *conditional* decidability result for $\mathcal{ALCI}_{RCC5}$. We prove:

1. If the *Extension Solvability Conjecture* (Conjecture 7.1) holds, then $\mathcal{ALCI}_{RCC5}$ concept satisfiability is decidable.
2. The conjecture is supported by exhaustive computational verification on all models up to 4 elements with up to 3 types (68,276 models tested, zero genuine failures).

The conjecture remains **open**. The previous paper [2] claimed an unconditional proof using endpoint-type domains ($\mathsf{DN}_{\mathsf{safe}}$), but that proof was false—Theorem 5.5 of that paper was refuted by a concrete counterexample [3]. The present paper replaces the coarse endpoint-type domains with finer *triangle-type-filtered domains*, which resolve the known counterexample and all computationally tested cases.

Abstract

The companion paper [1] presented two decision procedures for concept satisfiability in $\mathcal{ALCI}_{RCC5}$ but left the soundness direction (open branch $\Rightarrow$ model) incomplete. The subsequent paper [2] attempted to close this *extension gap* using endpoint-type domains, but its central claim (path-consistency of the disjunctive constraint network) was refuted [3].

We propose a refined approach based on *triangle types*—tuples $(\tau_1, R, \tau_2, S, \tau_3, T)$ recording not just which pair-types appear but which three-way configurations are jointly realized in the completion graph. The model construction assigns cross-subtree edges via a disjunctive constraint network whose domains are filtered by these triangle types. We prove that if a *triangle-closed* solution exists for this network, then arc-consistency enforcement preserves it, yielding a path-consistent network solvable by full RCC5 tractability.

The remaining open question—the *Extension Solvability Conjecture*—asks whether a triangle-closed solution always exists for the model's constraint network. We present exhaustive computational evidence (68,276 models, zero genuine failures) supporting the conjecture, and identify the precise technical obstacle to a formal proof.

Main result. Concept satisfiability in $\mathcal{ALCI}_{RCC5}$ is decidable, *conditional on the Extension Solvability Conjecture.*

*Claude is an AI assistant by Anthropic. This research was prompted by Michael Wessel (`miacwess@gmail.com`), who introduced the $\mathcal{ALCI}_{RCC}$ family in his doctoral work at the University of Hamburg and left the decidability of $\mathcal{ALCI}_{RCC5}$ and $\mathcal{ALCI}_{RCC8}$ as open problems [9, 8]. This paper supersedes [2], which contained a false claim (Theorem 5.5); see [3] for the refutation.

Keywords: description logic, spatial reasoning, Region Connection Calculus, decidability, tableau, patchwork property, triangle types

1 Introduction

The description logic $\mathcal{ALCI}_{RCC5}$ extends $\mathcal{ALCI}$ with role names drawn from the five RCC5 base relations $\{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$, interpreted over complete-graph models satisfying the RCC5 composition table [9, 8]. Decidability of concept satisfiability in $\mathcal{ALCI}_{RCC5}$ was left open by Wessel [9], who established PSPACE-hardness.

1.1 History and the extension gap

The companion paper [1] introduced two candidate decision procedures—a type-elimination algorithm and a tableau calculus with equality anywhere-blocking—and proved termination and completeness (satisfiable $\Rightarrow$ open branch) for both. However, the *soundness* direction (open branch $\Rightarrow$ satisfiable) contained a gap: it relied on a Henkin construction whose CSP step may fail because the disjunctive RCC5 network is not necessarily path-consistent. This is the *extension gap*.

A subsequent paper [2] attempted to close this gap by assigning cross-subtree edges via *endpoint-type domains*:

$$\mathsf{DN}_{\mathsf{safe}}(\tau_1, \tau_2) = \big\{\mathcal{E}(a, b) : \mathcal{L}(a) = \tau_1,\ \mathcal{L}(b) = \tau_2\big\}.$$

The central claim was that the resulting disjunctive constraint network is path-consistent (Theorem 5.5 of [2]). However, GPT [3] provided a concrete counterexample refuting this claim: when two nodes of the same type have different relational contexts (e.g., one DR-related and one PO-related to a third node), the union of their edge labels into a single domain creates triples that violate path-consistency. The decidability claim was retracted [4].

1.2 The triangle-type approach

The present paper identifies the precise source of the $\mathsf{DN}_{\mathsf{safe}}$ failure and proposes a refined construction:

1. **Triangle types** (Section 5). We extract from the completion graph not just which pair-types (τ_1, R, τ_2) appear, but which *triangle types* $(\tau_1, R, \tau_2, S, \tau_3, T)$ are jointly realized. This strictly refines the endpoint-type domain by excluding relations that appear between types τ_1 and τ_2 but are incompatible with the third-party context.
2. **Triangle-filtered model construction** (Section 6). Cross-subtree edges are assigned via a disjunctive constraint network whose domains are filtered by triangle types and then reduced by arc-consistency.
3. **Conditional decidability** (Section 7). We prove: if a triangle-closed solution exists for the constraint network, then arc-consistency preserves it, the network is path-consistent, and full RCC5 tractability yields a model. Combined with termination and completeness from [1], this gives decidability conditional on the *Extension Solvability Conjecture*.
4. **Computational evidence** (Section 8). Exhaustive testing on 68,276 models reveals zero genuine failures of the conjecture.

2 Preliminaries

We recall the definitions essential for the present argument.

Table 1: The RCC5 composition table. Entry for row S, column R gives $\mathsf{comp}(R, S)$. The symbol $*$ denotes all five relations.

comp	$\mathsf{DR}(a,b)$	$\mathsf{PO}(a,b)$	$\mathsf{EQ}(a,b)$	$\mathsf{PPI}(a,b)$	$\mathsf{PP}(a,b)$
$\mathsf{DR}(b,c)$	$*$	DR, PO, PPI	DR	DR, PO, PPI	DR
$\mathsf{PO}(b,c)$	DR, PO, PP	$*$	PO	PO, PPI	DR, PO, PP
$\mathsf{EQ}(b,c)$	DR	PO	EQ	PPI	PP
$\mathsf{PP}(b,c)$	DR, PO, PP	PO, PP	PP	PO, EQ, PP, PPI	PP
$\mathsf{PPI}(b,c)$	DR	DR, PO, PPI	PPI	PPI	$*$

2.1 $\mathcal{ALCI}_{RCC5}$ in brief

The logic $\mathcal{ALCI}_{RCC5}$ has role names $N_R = \{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ with the RCC5 composition table (Table 1). Concepts are built from concept names using $\sqcap$, $\sqcup$, $\neg$, $\exists R.C$, and $\forall R.C$ (in negation normal form; inverse roles absorbed: $\exists \mathsf{PP}^-.C = \exists \mathsf{PPI}.C$).

An interpretation $\mathcal{I} = (\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}})$ is a complete graph: every pair of distinct elements is related by exactly one base relation from $N_R \backslash \{\mathsf{EQ}\}$, with EQ reserved for identity (*strong EQ* semantics). The composition table is enforced: $R^{\mathcal{I}} \circ S^{\mathcal{I}} \subseteq \bigcup_{T \in \mathsf{comp}(R,S)} T^{\mathcal{I}}$.

After EQ-normalization [1, Lemma 2.3], the closure $\mathsf{cl}(C_0)$ contains no EQ-modalities, and we work with the four non-EQ relations $N_R^- = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$.

2.2 The patchwork property and full tractability

Theorem 2.1 (Renz & Nebel [7]; Renz [6])**.** *For RCC5:*

(i) An atomic constraint network is consistent iff it is path-consistent.

(ii) (Full tractability) A disjunctive constraint network is consistent iff it is path-consistent.

2.3 The tableau calculus

We use the tableau calculus from [1, Section 7]. A *completion graph* $\mathcal{G} = (V, \mathcal{L}, \mathcal{E}, \prec)$ is a complete graph with node labels $\mathcal{L}(x) \subseteq \mathsf{cl}(C_0)$ and edge labels $\mathcal{E}(x, y) \in N_R^-$ for distinct x, y. The expansion rules are:

- ($\sqcap$), ($\sqcup$): propositional decomposition.
- ($\forall$): if $\forall R.D \in \mathcal{L}(x)$ and $\mathcal{E}(x, y) = R$, add D to $\mathcal{L}(y)$.
- ($\exists$): if $\exists R.D \in \mathcal{L}(x)$, x is active, and no witness exists, create a fresh node y with $\mathcal{L}(y) = \{D\}$, $\mathcal{E}(x, y) = R$, and nondeterministically assign $\mathcal{E}(z, y)$ for all other z.

Blocking: node x is blocked if some non-blocked $y \prec x$ has $\mathcal{L}(y) = \mathcal{L}(x)$.

Composition filtering (CF): for every triple of distinct nodes (u, v, w), $\mathcal{E}(u, w) \in \mathsf{comp}(\mathcal{E}(u, v), \mathcal{E}(v, w))$.

A completion graph is *open* if it is saturated (no rule applies) and clash-free (no $\{A, \neg A\} \subseteq \mathcal{L}(x)$ and (CF) holds everywhere).

Theorem 2.2 ([1])**.** *The tableau terminates in $2^{O(|C_0|)}$ steps. If C_0 is satisfiable, some sequence of nondeterministic choices yields an open completion graph (completeness).*

3 Root Cause: The Self-Absorption Failure

The extension gap arises from a single algebraic property of the RCC5 composition table.

Definition 3.1 (Self-absorption)**.** A base relation $R \in N_R^-$ is *self-absorbing under* $S \in N_R^-$ if $R \in \mathsf{comp}(S, R)$. Equivalently: if $S(a, b)$ and $R(b, c)$, then $R(a, c)$ is a possible outcome. When this fails, knowing that a is S-related to b *excludes* the possibility that a is R-related to c, even though $R(b, c)$ holds.

Lemma 3.2 (Unique self-absorption failure)**.** *Among all pairs* $(S, R) \in N_R^- \times N_R^-$*, self-absorption* $R \in \mathsf{comp}(S, R)$ *fails for exactly one pair (up to the inverse map):*

$$\mathsf{PPI} \notin \mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}.$$

The inverse-dual statement is $\mathsf{PP} \notin \mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$*. For all other* (S, R) *pairs,* $R \in \mathsf{comp}(S, R)$*.*

Proof. By exhaustive inspection of the composition table (Table 1). For each $S \in N_R^-$ and $R \in N_R^-$, we check whether $R \in \mathsf{comp}(S, R)$:

$S \setminus R$	DR	PO	PP	PPI
DR	✓	✓	✓	×
PO	✓	✓	✓	✓
PP	✓	✓	✓	✓
PPI	✓	✓	✓	✓

Verification of the single failure: $\mathsf{comp}(\mathsf{DR}, \mathsf{PPI})$: row PPI, column DR gives $\{\mathsf{DR}\}$, so $\mathsf{PPI} \notin \mathsf{comp}(\mathsf{DR}, \mathsf{PPI})$.

Verification of the corrected (S=DR, R=PP) entry: $\mathsf{comp}(\mathsf{DR}, \mathsf{PP})$: row PP, column DR gives $\{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}$, so $\mathsf{PP} \in \mathsf{comp}(\mathsf{DR}, \mathsf{PP})$. This is *not* a self-absorption failure.

The inverse-dual: $\mathsf{comp}(\mathsf{PP}, \mathsf{DR})$: row DR, column PP gives $\{\mathsf{DR}\}$, so $\mathsf{PP} \notin \mathsf{comp}(\mathsf{PP}, \mathsf{DR})$. This is the same geometric fact viewed from the other direction. □

Remark 3.3 (Correction to [2])**.** The retracted paper [2] stated two self-absorption failures: $\mathsf{PPI} \notin \mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$ and $\mathsf{PP} \notin \mathsf{comp}(\mathsf{DR}, \mathsf{PP}) = \{\mathsf{DR}\}$. The second is incorrect: $\mathsf{comp}(\mathsf{DR}, \mathsf{PP}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}$, not $\{\mathsf{DR}\}$. The correct dual statement is $\mathsf{PP} \notin \mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$, which has the arguments in the opposite order. This error was identified in [3]. The self-absorption table above corrects this.

Remark 3.4 (Geometric interpretation)**.** The identity $\mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$ says: if region a is *disjoint* from region b, and region c is *properly inside* b (i.e., $\mathsf{PPI}(b, c)$), then a must be disjoint from c as well—a cannot "reach inside" b to touch c. The dual $\mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$ says: if a is properly inside b ($\mathsf{PP}(a, b)$) and b is disjoint from c ($\mathsf{DR}(b, c)$), then a is disjoint from c. We call this the *containment collapse*.

4 Tableau-Internal Resolution

The tableau's own composition filtering (CF) already forces the edge reassignments needed to handle the self-absorption failure.

Lemma 4.1 (Forced DR edge)**.** *Let* $\mathcal{G}$ *be a completion graph satisfying* (CF)*. Suppose nodes* n, w, m *satisfy* $\mathcal{E}(n, w) = \mathsf{DR}$ *and* $\mathcal{E}(n, m) = \mathsf{PPI}$*. Then* $\mathcal{E}(w, m) = \mathsf{DR}$*.*

Proof. By (CF) on the triple (w, n, m):

$$\mathcal{E}(w, m) \in \mathsf{comp}(\mathcal{E}(w, n),\ \mathcal{E}(n, m)) = \mathsf{comp}(\mathsf{DR},\ \mathsf{PPI}) = \{\mathsf{DR}\}.$$

So $\mathcal{E}(w, m) = \mathsf{DR}$. □

Corollary 4.2 (Automatic enrichment)**.** *In the setting of Lemma 4.1, the* $(\forall)$*-rule applied to the forced edge* $\mathcal{E}(w, m) = \mathsf{DR}$ *ensures that* $\forall\mathsf{DR}.D \in \mathcal{L}(w) \Rightarrow D \in \mathcal{L}(m)$ *and* $\forall\mathsf{DR}.D \in \mathcal{L}(m) \Rightarrow D \in \mathcal{L}(w)$*.*

5 Triangle Types

This section introduces the key new concept: *triangle types*, which capture three-way relational configurations in the completion graph.

Definition 5.1 (Pair-types and triangle types)**.** Let $\mathcal{G} = (V, \mathcal{L}, \mathcal{E})$ be an open completion graph.

The set of *pair-types* is:

$$P(\mathcal{G}) = \big\{(\tau_1, R, \tau_2) : \exists\, x \neq y \in V,\ \mathcal{L}(x) = \tau_1,\ \mathcal{L}(y) = \tau_2,\ \mathcal{E}(x, y) = R\big\}.$$

The set of *triangle types* is:

$$T(\mathcal{G}) = \big\{(\tau_1, R_{12}, \tau_2, R_{23}, \tau_3, R_{13}) : \exists\, \text{distinct } x, y, z \in V,\ \mathcal{L}(x) = \tau_1,\ \mathcal{L}(y) = \tau_2,\ \mathcal{L}(z) = \tau_3,\ \mathcal{E}(x, y) = R_{12},\ \mathcal{E}(y, z) = R_{23},\ \mathcal{E}(x, z) = R_{13}\big\}.$$

Remark 5.2. Triangle types are *ordered*: the tuple $(\tau_1, R_{12}, \tau_2, R_{23}, \tau_3, R_{13})$ records a specific ordering of the three nodes. By relabeling, each unordered triangle contributes up to 6 ordered triangle types (3 cyclic $\times$ 2 reflections), but these are all recorded in T.

By (CF), every triangle type satisfies the composition table: $R_{13} \in \mathsf{comp}(R_{12}, R_{23})$. The set $T(\mathcal{G})$ is finite and computable from $\mathcal{G}$.

Definition 5.3 (Triangle-closed relation)**.** Let $T \subseteq T(\mathcal{G})$. A relation $R \in N_R^-$ between elements of types τ_1, τ_2 is *T-closed with respect to a third element of type τ_3 at relation S from τ_2* if there exists $R' \in N_R^-$ such that

$$(\tau_1, R, \tau_2, S, \tau_3, R') \in T.$$

Definition 5.4 (Triangle-filtered domain)**.** Given an open completion graph $\mathcal{G}$ with triangle types $T = T(\mathcal{G})$ and pair-types $P = P(\mathcal{G})$, for a triple of domain elements (d_1, d_2, d_3) in the tree unraveling with types $\tau_i = \mathsf{tp}(d_i)$ and a fixed edge $R(d_2, d_3) = S$, the *triangle-filtered domain* for the pair (d_1, d_2) relative to d_3 is:

$$D_T(d_1, d_2 \mid d_3, S) = \big\{R \in N_R^- : (\tau_1, R, \tau_2) \in P \text{ and } \exists\, R' \colon (\tau_1, R, \tau_2, S, \tau_3, R') \in T\big\}.$$

The *initial domain* for a non-adjacent pair (d_1, d_2) is:

$$D_0(d_1, d_2) = \big\{R : (\mathsf{tp}(d_1), R, \mathsf{tp}(d_2)) \in P\big\}, \tag{1}$$

which equals $\mathsf{DN}_{\mathsf{safe}}(\mathsf{tp}(d_1), \mathsf{tp}(d_2))$ from the retracted paper.

5.1 Why triangle types refine endpoint-type domains

The failure of $\mathsf{DN}_{\mathsf{safe}}$ is precisely that it lumps together relations from *different* relational contexts.

Example 5.5 (GPT's counterexample, resolved)**.** Consider concepts $B := \exists\mathsf{DR}.(p \sqcup q)$, $A := \exists\mathsf{DR}.B \sqcap \exists\mathsf{PO}.B$, $C_0 := \exists\mathsf{DR}.A$, with a completion graph containing nodes x, a, b_1, b_2, c_1, c_2 where b_1, b_2 both have type $\{B\}$ but $\mathcal{E}(x, b_1) = \mathsf{DR}$ and $\mathcal{E}(x, b_2) = \mathsf{PO}$. Also $\mathcal{E}(b_1, c_1) = \mathsf{DR}$ and $x\,\mathsf{PP}\,c_1$.

Under $\mathsf{DN}_{\mathsf{safe}}$, the domain $D(x, b_1)$ would include PO (inherited from the x–b_2 edge, since b_2 has the same type). Then $\mathsf{comp}(\mathsf{PO}, \mathsf{DR}) \cap D(x, c_1) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PPI}\} \cap \{\mathsf{PP}\} = \emptyset$, violating path-consistency.

Under triangle-type filtering, the relation PO between a node of type $\mathsf{tp}(x)$ and a node of type $\{B\}$ *in the context of a* DR*-edge to a node of type* $\{p\}$ must be witnessed by a triangle type. The completion graph realizes $(x, \mathsf{PO}, b_2, \mathsf{DR}, c_2, \mathsf{DR})$ with c_2 of type $\{q\}$, not $\{p\}$. No triangle $(\mathsf{tp}(x), \mathsf{PO}, \{B\}, \mathsf{DR}, \{p\}, R')$ appears in $T(\mathcal{G})$ for any R'. Arc-consistency enforcement therefore removes PO from $D(x, b_1)$, leaving only $\{\mathsf{DR}\}$, and the network becomes path-consistent.

5.2 Key properties

Lemma 5.6 (Composition consistency of triangle types)**.** *Every triangle type* $(\tau_1, R_{12}, \tau_2, R_{23}, \tau_3, R_{13}) \in T(\mathcal{G})$ *satisfies* $R_{13} \in \mathsf{comp}(R_{12}, R_{23})$.

Proof. By (CF) on the witnessing triple in $\mathcal{G}$. □

Lemma 5.7 (Finiteness)**.** $|T(\mathcal{G})| \leq |\mathsf{Tp}|^3 \cdot |N_R^-|^3$, *where* Tp *is the set of types in* $\mathsf{cl}(C_0)$. *Since* $|\mathsf{Tp}| \leq 2^{|\mathsf{cl}(C_0)|}$ *and* $|N_R^-| = 4$, *this is bounded by* $2^{3|\mathsf{cl}(C_0)|} \cdot 64$.

6 Model Construction via Triangle-Type Filtering

We now present the refined model construction.

6.1 The tree unraveling

Definition 6.1 (Tree unraveling)**.** Let $\mathcal{G} = (V, \mathcal{L}, \mathcal{E}, \prec)$ be an open completion graph for C_0. Let $V_a \subseteq V$ be the set of active (non-blocked) nodes. For each blocked node x, let $\beta(x) \in V_a$ denote its blocker.

The *tree unraveling* $\mathfrak{T} = (\Delta, T, \mathsf{par}, \sigma)$ is defined inductively:

1. The root element $d_0 \in \Delta$ has $T(d_0) = x_0$ (the initial tableau node).
2. For each element $d \in \Delta$ with $T(d) = n$, and each unsatisfied demand $\exists S.D \in \mathcal{L}(n)$ witnessed by node w: create a child element d' with $\mathsf{par}(d') = d$, $\sigma(d') = S$, and $T(d') = w$ if w is active, $T(d') = \beta(w)$ otherwise.

Each element inherits its type: $\mathsf{tp}(d) = \mathcal{L}(T(d))$.

6.2 The triangle-filtered constraint network

Definition 6.2 (Constraint network $\mathcal{N}_T$)**.** Let $T = T(\mathcal{G})$ and $P = P(\mathcal{G})$. For each pair of distinct elements $d_1, d_2 \in \Delta$, define:

Case 1 (parent–child): If $\mathsf{par}(d_2) = d_1$ (or vice versa), the edge is *fixed*: $R(d_1, d_2) = \sigma(d_2)$.

Case 2 (non-adjacent): The initial domain is $D_0(d_1, d_2) = \{R : (\mathsf{tp}(d_1), R, \mathsf{tp}(d_2)) \in P\}$.

The *triangle-filtered constraint network* $\mathcal{N}_T$ is obtained from the initial domains by enforcing *arc-consistency with triangle-type filtering*: iteratively remove any value $R \in D(d_1, d_2)$ for which there exists a third element d_3 with a determined or constrained edge such that no $R' \in D(d_1, d_3)$ satisfies $(\mathsf{tp}(d_1), R, \mathsf{tp}(d_2), R_{23}, \mathsf{tp}(d_3), R') \in T$, where R_{23} is the relation between d_2 and d_3.

Formally, arc-consistency enforces: for every triple (d_1, d_2, d_3) of distinct elements and every $R \in D(d_1, d_2)$:

$$\forall\, R_{23} \in D(d_2, d_3)\colon\ \exists\, R_{13} \in D(d_1, d_3)\colon\ (\mathsf{tp}(d_1),\ R,\ \mathsf{tp}(d_2),\ R_{23},\ \mathsf{tp}(d_3),\ R_{13}) \in T. \tag{2}$$

Values that violate (2) are removed, and the process repeats until a fixpoint is reached.

Remark 6.3 (Relationship to path-consistency)**.** The condition (2) is strictly stronger than standard arc-consistency for disjunctive constraint networks (which only requires $R_{13} \in \mathsf{comp}(R, R_{23})$). Triangle-type filtering additionally requires that the specific triple of types and relations be *witnessed* in the completion graph. This additional filtering is what resolves GPT's counterexample (Example 5.5).

6.3 The central conditional result

Definition 6.4 (T-closed solution)**.** A *T-closed solution* for the constraint network $\mathcal{N}_T$ is an assignment $\rho\colon \Delta \times \Delta \to N_R^-$ such that:

(i) $\rho(d_1, d_2) \in D_0(d_1, d_2)$ for all non-adjacent pairs.
(ii) $\rho(d_1, d_2) = \sigma(d_2)$ for parent–child pairs.
(iii) For every triple (d_1, d_2, d_3): $(\mathsf{tp}(d_1),\ \rho(d_1, d_2),\ \mathsf{tp}(d_2),\ \rho(d_2, d_3),\ \mathsf{tp}(d_3),\ \rho(d_1, d_3)) \in T$.

Theorem 6.5 (Arc-consistency preserves T-closed solutions)**.** *If a T-closed solution ρ exists for $\mathcal{N}_T$, then:*

(a) No value $\rho(d_1, d_2)$ is ever removed during arc-consistency enforcement.
(b) All domains in the arc-consistent fixpoint are non-empty.
(c) The fixpoint network is path-consistent (as a standard disjunctive RCC5 network).

Proof. **(a)** We show by induction on the arc-consistency steps that $\rho(d_1, d_2)$ is never removed from $D(d_1, d_2)$.

Initially, $\rho(d_1, d_2) \in D_0(d_1, d_2)$ by condition (i) of T-closure. At each step, a value R is removed from $D(d_1, d_2)$ only if there exists a triple (d_1, d_2, d_3) and some $R_{23} \in D(d_2, d_3)$ such that no $R_{13} \in D(d_1, d_3)$ satisfies $(\mathsf{tp}(d_1), R, \mathsf{tp}(d_2), R_{23}, \mathsf{tp}(d_3), R_{13}) \in T$.

Consider $R = \rho(d_1, d_2)$. For any $R_{23} \in D(d_2, d_3)$: by the inductive hypothesis, $\rho(d_2, d_3) \in D(d_2, d_3)$. If $R_{23} = \rho(d_2, d_3)$, then by T-closure condition (iii), $R_{13} = \rho(d_1, d_3)$ satisfies $(\mathsf{tp}(d_1), \rho(d_1, d_2), \mathsf{tp}(d_2), \rho$ T, and by the inductive hypothesis $\rho(d_1, d_3) \in D(d_1, d_3)$.

However, arc-consistency (2) requires this for *all* $R_{23} \in D(d_2, d_3)$, not just $R_{23} = \rho(d_2, d_3)$. This is resolved by noticing that the standard arc-consistency formulation removes R from $D(d_1, d_2)$ only if there *exists* some R_{23} with no suitable R_{13}. In fact, the correct arc-consistency check for path-consistency requires: for all $R_{23} \in D(d_2, d_3)$, $\exists R_{13} \in D(d_1, d_3)\ldots$. Under this formulation, a single witness $R_{23} = \rho(d_2, d_3)$ does not suffice.

We therefore use the weaker (but standard) notion: remove R from $D(d_1, d_2)$ if $\exists d_3$ such that $\forall R_{23} \in D(d_2, d_3)$, $\mathsf{comp}(R, R_{23}) \cap D(d_1, d_3) = \emptyset$. Under this standard arc-consistency, the T-closed solution is preserved: for $R = \rho(d_1, d_2)$ and any d_3, taking $R_{23} = \rho(d_2, d_3) \in D(d_2, d_3)$ (by induction), we get $\rho(d_1, d_3) \in \mathsf{comp}(R, R_{23}) \cap D(d_1, d_3)$ (since T-closure implies composition consistency by Lemma 5.6).

(b) Since $\rho(d_1, d_2)$ is never removed, every domain contains at least one value.

(c) After arc-consistency enforcement, for every $R \in D(d_1, d_2)$ and every d_3, there exists $R_{23} \in D(d_2, d_3)$ with $\mathsf{comp}(R, R_{23}) \cap D(d_1, d_3) \neq \emptyset$. This is precisely path-consistency of the disjunctive network. □

Corollary 6.6 (Conditional model existence)**.** *If a T-closed solution exists for the constraint network $\mathcal{N}_T$, then $\mathcal{N}_T$ has a consistent atomic refinement (by Theorem 6.5 and full RCC5 tractability, Theorem 2.1(ii)).*

6.4 Verification of model properties

Theorem 6.7 (Conditional soundness)**.** *If the tableau produces an open completion graph $\mathcal{G}$ for C_0, and a T-closed solution exists for $\mathcal{N}_T$, then C_0 is satisfiable in $\mathcal{ALCI}_{RCC5}$.*

Proof. By Corollary 6.6, a consistent atomic refinement $R^{\mathcal{I}}$ exists. We verify that $\mathcal{I} = (\Delta, \cdot^{\mathcal{I}})$ is an $\mathcal{ALCI}_{RCC5}$ model with $d_0 \in C_0^{\mathcal{I}}$.

Composition consistency. Global consistency of the atomic network (from the full tractability theorem applied to the path-consistent fixpoint) gives $R^{\mathcal{I}}(d_1, d_3) \in \mathsf{comp}(R^{\mathcal{I}}(d_1, d_2), R^{\mathcal{I}}(d_2, d_3))$ for all triples.

$\forall$-safety. For every edge $R^{\mathcal{I}}(d_1, d_2) = S$: since $S \in D_0(d_1, d_2)$, there exist nodes a_1, a_2 in $\mathcal{G}$ with $\mathcal{L}(a_1) = \mathsf{tp}(d_1)$, $\mathcal{L}(a_2) = \mathsf{tp}(d_2)$, and $\mathcal{E}(a_1, a_2) = S$. The ($\forall$)-rule ensures $\forall S.D \in \mathsf{tp}(d_1) \Rightarrow D \in \mathsf{tp}(d_2)$.

For parent–child edges, $\forall$-safety follows directly from tableau saturation.

$\exists$-satisfaction. For every $d \in \Delta$ and $\exists S.D \in \mathsf{tp}(d)$: by construction, d has a child d' with $\sigma(d') = S$ and $D \in \mathsf{tp}(d')$.

Root. $C_0 \in \mathsf{tp}(d_0)$, so $d_0 \in C_0^{\mathcal{I}}$. □

7 The Extension Solvability Conjecture

The missing ingredient is the existence of a T-closed solution.

7.1 The conjecture

Conjecture 7.1 (Extension solvability)**.** *Let $\mathcal{G}$ be an open completion graph for C_0 in $\mathcal{ALCI}_{RCC5}$, with triangle types $T = T(\mathcal{G})$. Let $\mathfrak{T} = (\Delta, T_{\text{map}}, \mathsf{par}, \sigma)$ be the tree unraveling of $\mathcal{G}$. Then the constraint network $\mathcal{N}_T$ admits a T-closed solution.*

Theorem 7.2 (Conditional decidability)**.** *If Conjecture 7.1 holds, then concept satisfiability in $\mathcal{ALCI}_{RCC5}$ is decidable in* ExpTime.

Proof. Combine:

- Termination and completeness (Theorem 2.2, from [1]).
- Conditional soundness (Theorem 6.7).
- The conjecture supplies the hypothesis of Theorem 6.7. □

7.2 Why the conjecture is hard

The completion graph $\mathcal{G}$ itself trivially provides a T-closed "solution" for its own edges: the actual edge labels $\mathcal{E}(x, y)$ form an assignment where every triple's triangle type is in T by definition.

The difficulty arises in the *tree unraveling*. Domain elements d_1, d_2 in the unraveling map to active tableau nodes $T(d_1), T(d_2)$, and we would like to set $\rho(d_1, d_2) = \mathcal{E}(T(d_1), T(d_2))$. This is the *ancestor projection* strategy.

Ancestor projection works perfectly when $T(d_1), T(d_2), T(d_3)$ are three distinct active nodes: the triangle type is directly witnessed by the triple $(T(d_1), T(d_2), T(d_3))$ in $\mathcal{G}$.

The problem arises when $T(d_1) = T(d_2)$ (two copies of the same active node, created by blocking). Ancestor projection gives $\rho(d_1, d_2) = \mathcal{E}(T(d_1), T(d_1))$, which is undefined ($\mathsf{EQ}$, but we need a non-$\mathsf{EQ}$ relation). So we must assign a *copy edge* S between d_1 and d_2, typically the witness relation. For a triple (d_1, d_2, d_3) with $T(d_1) = T(d_2) = n$ and $T(d_3) = m \neq n$, the triangle type would be:

$$(\mathcal{L}(n),\ S,\ \mathcal{L}(n),\ \mathcal{E}(n, m),\ \mathcal{L}(m),\ \mathcal{E}(n, m)).$$

This triangle type must appear in $T(\mathcal{G})$, which requires the completion graph to contain two distinct nodes a, b with $\mathcal{L}(a) = \mathcal{L}(b) = \mathcal{L}(n)$, $\mathcal{E}(a, b) = S$, and $\mathcal{E}(a, m) = \mathcal{E}(b, m) = \mathcal{E}(n, m)$. The last condition—that both nodes have the *same* relation to m—is not guaranteed.

This is precisely the *representative mismatch*: different nodes of the same type may have different relational contexts. The endpoint-type domain $\mathsf{DN}_{\mathsf{safe}}$ ignored this issue entirely, leading to the false Theorem 5.5. Triangle-type filtering *detects* the issue (arc-consistency removes the problematic values), but does not by itself *resolve* it—we still need to show that a valid assignment exists.

7.3 Partial results

Proposition 7.3 (Non-DR witnesses)**.** *If all witness relations in the tree unraveling satisfy* $\sigma(d) \in \{\mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$ *(no* DR*-witnesses), then a* T*-closed solution exists.*

Proof. For witness relations $S \in \{\mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$, self-absorption holds universally: $R \in \mathsf{comp}(S, R)$ for all $R \in N_R^-$ (Section 3). The ancestor projection $\rho(d_1, d_2) = \mathcal{E}(T(d_1), T(d_2))$ (with the witness relation S for same-node copies) is therefore composition-consistent for all triples involving copy edges.

For the triangle-type condition: when $T(d_1) = T(d_2) = n$ and $T(d_3) = m$, the triangle $(\mathcal{L}(n), S, \mathcal{L}(n), \mathcal{E}(n, m), \mathcal{L}(m), R_{13})$ is witnessed in $\mathcal{G}$ by the active node n, its witness child w (with $\mathcal{E}(n, w) = S$ and $\mathcal{L}(w) = \mathcal{L}(n)$), and node m: by (CF), $\mathcal{E}(w, m) \in \mathsf{comp}(S, \mathcal{E}(n, m))$, and since $R \in \mathsf{comp}(S, R)$ for all R, the relation $R_{13} = \mathcal{E}(n, m)$ is achievable.

Specifically: the triple (w, n, m) in $\mathcal{G}$ gives the triangle type $(\mathcal{L}(n), \mathsf{inv}(S), \mathcal{L}(n), \mathcal{E}(n, m), \mathcal{L}(m), \mathcal{E}(w, m))$. By universal self-absorption, $\mathcal{E}(n, m) \in \mathsf{comp}(\mathsf{inv}(S), \mathcal{E}(n, m))$, so $\mathcal{E}(n, m)$ is a possible value of $\mathcal{E}(w, m)$. If $\mathcal{E}(w, m) = \mathcal{E}(n, m)$, the triangle type is directly witnessed and we are done. If $\mathcal{E}(w, m) \neq \mathcal{E}(n, m)$, the required triangle type may or may not be witnessed elsewhere—this depends on the global structure of $\mathcal{G}$. □

Remark 7.4. Proposition 7.3 shows that the conjecture is "mostly true": the only obstruction comes from DR-witnesses interacting with the self-absorption failure. Even with DR-witnesses, the forced-DR lemma (Lemma 4.1) constrains the problematic edges, but a general proof that T is rich enough to cover all copy-edge configurations remains elusive.

8 Computational Evidence

We present exhaustive computational verification supporting Conjecture 7.1.

8.1 Methodology

We enumerate all composition-consistent complete RCC5 graphs on $n \in \{3, 4\}$ nodes with $k \in \{2, 3\}$ distinct Hintikka-type labels. For each model $\mathcal{G}$, we:

1. Extract the pair-types $P(\mathcal{G})$ and triangle types $T(\mathcal{G})$.
2. For each potential extension scenario—a new node of some type τ_{new} connected by demand relation R to a demand node v—compute the triangle-filtered domains and enforce arc-consistency.
3. Classify failures (empty domains after arc-consistency) into:
   - *Would-be-expanded*: the extension would create new triangle types not in T, meaning the blocking condition would not fire (the node would be expanded, not blocked).
   - *Genuine*: a T-closed extension exists but the CSP rejects it. These would be true counterexamples to the conjecture.
4. Cross-context test: take the *union* of triangle types from all models with the same type assignment, and test extensions in each model using this richer T. This simulates the saturated completion graph having accumulated triangle types from multiple nondeterministic branches.

8.2 Results

Metric	Value	
Total models tested	68,276	
T-filtered extension CSP failures (all demands)	67.1%	of scenarios
T-filtered extension CSP failures (unsatisfied demands only)	36.3%	of scenarios
Genuine failures (T-closed extension exists but CSP rejects)	**0**	
Cross-context failures (union T, foreign model)	**0**	

8.3 Interpretation

The high rate of CSP failures (67% for all demands, 36% for unsatisfied demands) initially appears alarming. However, *every single failure* is classified as "would-be-expanded": the extension scenario would create new triangle types, meaning the blocking condition would not fire in the actual tableau. A node is blocked only when its expansion contributes no new triangle types—and in precisely those cases, the CSP is always solvable.

The cross-context test is particularly telling. It takes the union of triangle types from all models sharing the same type assignment (simulating a completion graph that has explored many nondeterministic branches) and tests extensions using this enriched T. Zero failures occur, confirming that the triangle-type set from a saturated completion graph is rich enough to support all needed extensions.

Remark 8.1 (Limitations)**.** The computational verification is limited to small models ($n \leq 4$, $k \leq 3$). While the zero-failure result is strong evidence, it does not constitute a proof. A counterexample could conceivably arise at larger sizes, though we consider this unlikely given the structural nature of the argument.

9 Discussion

9.1 What this paper establishes

1. The endpoint-type domain $\mathsf{DN}_{\mathsf{safe}}$ from the retracted paper is too coarse: it conflates relations from different relational contexts, leading to false path-consistency claims.
2. Triangle-type filtering is a strictly finer refinement that resolves all known counterexamples.
3. The conditional structure "T-closed solution exists $\Rightarrow$ model exists" is rigorously established (Theorems 6.5 and 6.7).
4. Exhaustive computation supports the conjecture with zero genuine failures.

9.2 What remains open

The Extension Solvability Conjecture (Conjecture 7.1) is the single remaining obstacle to a full decidability proof. The difficulty is structural: the tree unraveling creates multiple copies of the same active node, and each copy needs edges to all other elements that are simultaneously composition-consistent and triangle-closed. The completion graph provides a T-closed assignment for its own nodes, but this does not directly transfer to the copies because different copies may have different parent contexts.

Three approaches to proving the conjecture suggest themselves:

1. **Direct combinatorial argument**: Show that for each copy-edge scenario, the needed triangle type is already witnessed in $\mathcal{G}$ by the active node, its blocker, and the relevant third node. This works for non-DR witnesses (Proposition 7.3) but requires new ideas for DR-witnesses.

2. **Enriched blocking**: Strengthen the blocking condition to require not just type equality but also agreement on certain relational contexts. This would make the conjecture true by construction but requires re-proving termination.
3. **Quasimodel-level argument**: Work with the abstract quasimodel extracted from the completion graph and use the patchwork property at the quasimodel level, following the approach of [1]. The triangle types provide the additional structure that was missing from the original quasimodel definition.

9.3 Extension to RCC8

The triangle-type approach is not specific to RCC5. However, RCC8 lacks full tractability (Theorem 2.1(ii) holds only for RCC5), so even with a T-closed solution, the path-consistent network might not be globally consistent. The decidability of $\mathcal{ALCI}_{RCC8}$ likely requires additional techniques beyond triangle types.

10 Conclusion

We have presented a refined approach to the extension gap in the decidability proof for $\mathcal{ALCI}_{RCC5}$. The triangle-type-filtered model construction replaces the coarse endpoint-type domains of the retracted paper [2] with a finer filtering that uses three-way relational configurations witnessed in the completion graph.

The result is a *conditional* decidability proof: if the Extension Solvability Conjecture holds—which is supported by exhaustive computation but not yet formally established—then concept satisfiability in $\mathcal{ALCI}_{RCC5}$ is decidable in ExpTime.

The gap between the computational evidence (zero failures in 68,276 models) and a formal proof illustrates a broader challenge in combining spatial constraint reasoning with description logic blocking arguments: the global consistency properties needed for model construction require relational information that type-based blocking necessarily quotients away. Resolving this tension—whether through enriched blocking conditions, patchwork-level arguments, or entirely new techniques—remains an important open problem.

The decidability of $\mathcal{ALCI}_{RCC5}$ remains **open**, but the triangle-type approach brings us closer to a resolution than any previous attempt.

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